Circuit Simulation Lab:32
Design of Sequence Detector

Introduction

Design a sequential circuit (sequence detector) has a serial input X and an output Z. When the circuit receives a sequence of four consecutive ones (1111) on input X, it produces an output Z = 1 at the same time as the fourth input. At all other times, the output Z = 0. After the circuit receives a successful input sequence of 1111 (which produces Z = 1), a new input sequence of 1111 must be received again to produce Z = 1. (That is, the successful input sequences of 1111 must be non-overlapping.)

An example input/output sequence is shown below.

X: 0 0 1 1 1 0 1 1 0 0 1 1 1 1 0 0 1 1 1 1 1 1 1 1 1 1 0 0 . . .

Z: 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 0 1 0 0 0 0 . . .

Design this sequence detector procedurally, using D flip-flops. Also describe the FSM and verify the same with a Test Bench

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado